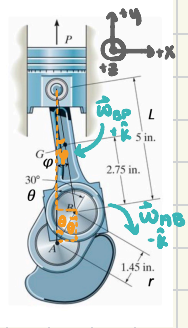
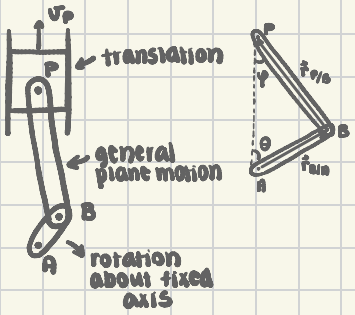
Project 1: Part B

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Step 1: Derive Analytical Relationship

Velocity of piston as function of angular rate of crank AB





For general plane motion:

$$\vec{v}_B = \vec{v}_A + \vec{\omega}_{AB} \times \vec{r}_{B/A}$$

In this case, we have

$$\vec{v}_P = \vec{v}_B + \vec{\omega}_{BP} \times \vec{r}_{P/B}$$

Since AB is rotating about a fixed axis, we have

$$\vec{v}_B = \vec{\omega}_{AB} \times \vec{r}_{B/A}$$

So

$$\vec{v}_P = \vec{\omega}_{AB} \times \vec{r}_{B/A} + \vec{\omega}_{BP} \times \vec{r}_{P/B}$$

$$\vec{v}_P = \underset{-\hat{k}}{\vec{\omega}_{AB}} \times r(\sin\theta\,\hat{i} + \cos\theta\,\hat{j}) + \underset{+\hat{k}}{\vec{\omega}_{BP}} \times L(-\sin\varphi\,\hat{i} + \cos\varphi\,\hat{j})$$

$$\vec{v}_P = \omega_{AB} r(-\sin\theta\,\hat{j} + \cos\theta\,\hat{i}) + \omega_{BP} L(-\sin\varphi\,\hat{j} - \cos\varphi\,\hat{i})$$

$$\vec{v}_P = (\omega_{AB} r\cos\theta - \omega_{BP} L\cos\varphi)\hat{i} + (-\omega_{AB} r\sin\theta - \omega_{BP} L\sin\varphi)\hat{j}$$

Horizontal component should be zero since piston only moves vertically, so

$$\omega_{AB} r\cos\theta - \omega_{BP} L\cos\varphi = 0$$

$$\omega_{AB} r\cos\theta = \omega_{BP} L\cos\varphi$$

$$\cos\varphi = \frac{\omega_{AB} r\cos\theta}{\omega_{BP} L} \Rightarrow \omega_{BP} = \frac{\omega_{AB} r\cos\theta}{L\cos\varphi} \quad *$$

$$\vec{v}_P = (\omega_{AB} r\cos\theta - \omega_{BP} L \frac{\omega_{AB} r\cos\theta}{\omega_{BP} L})\hat{i} + (-\omega_{AB} r\sin\theta - \omega_{BP} L\sin\varphi)\hat{j}$$

$$\vec{v}_P = (\omega_{AB} r\cos\theta - \omega_{AB} r\cos\theta)\hat{i} + (-\omega_{AB} r\sin\theta - \omega_{BP} L\sin\varphi)\hat{j}$$

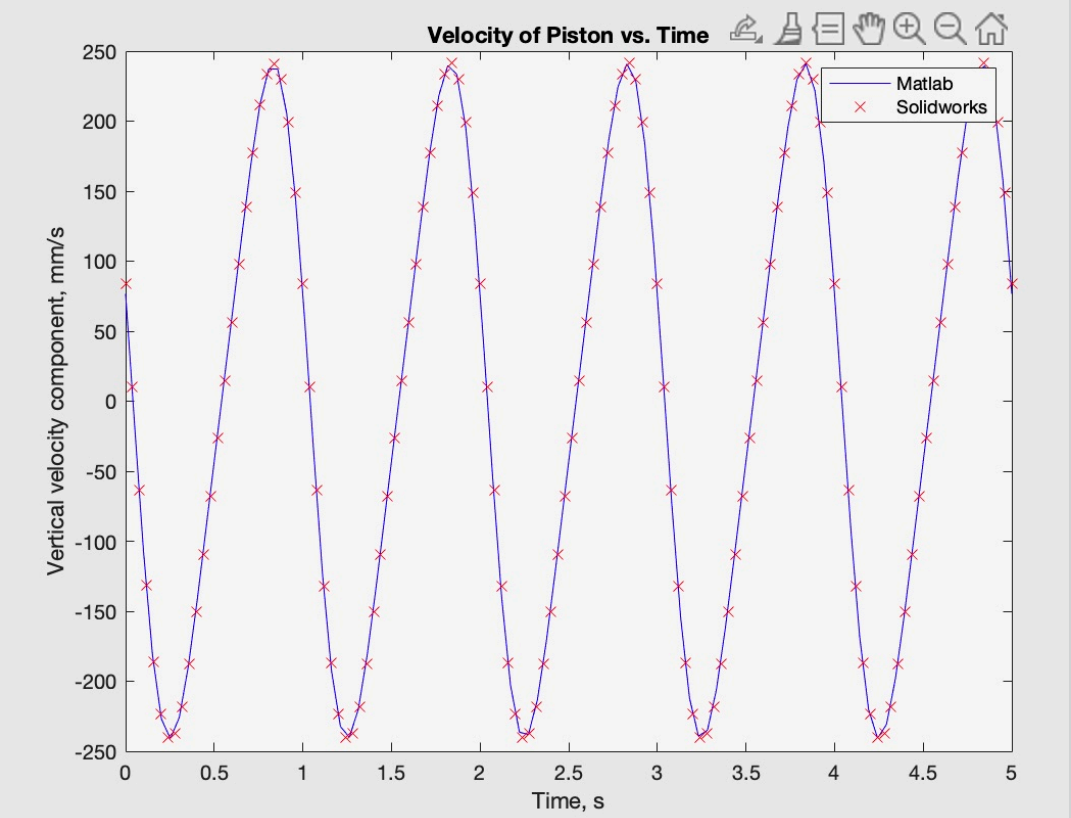
$$v_P = -\omega_{AB} r\sin\theta - \omega_{BP} L\sin\varphi$$

$$v_P = -\omega_{AB} r\sin\theta - \frac{\omega_{AB} r\cos\theta}{L\cos\varphi} L\sin\varphi \quad \text{plug in } *$$

$$v_P = -\omega_{AB} r\sin\theta - \omega_{AB} r\cos\theta\tan\varphi$$

$$v_P = -\omega_{AB} r(\sin\theta + \cos\theta\tan\varphi)$$

Step 2 and 3: MATLAB and SolidWorks piston velocity comparison



Matlab Code

```
%MATLAB
clc
clear all
close all

% Crank geometry
t = linspace(0,5);          % time, s
thetaDot = 2*pi;            % crank angular rate, rad/s
theta0 = pi/2;              % initial angle, rad
theta = thetaDot*t+theta0;  % crank angle, rad
r = 1.45*25.4;                  % crank radius, mm
L = 5*25.4;                     % rod length, mm

% Rod geometry
phi = asin(r/L*sin(theta));                  % rad
v = -thetaDot*r*(sin(theta) + cos(theta).*tan(phi)); % mm/s

% Load SolidWorks data
SWmatrix = readmatrix('Velocity Piston-1.csv');
SW_t = SWmatrix(:,1);
SW_v = SWmatrix(:,2);

% Plot
figure(1); clf;
plot(t,v,'b',SW_t,SW_v,'rx');
xlabel('Time, s');
ylabel('Vertical velocity component, mm/s');
legend('Matlab','Solidworks')
title('Velocity of Piston vs. Time')


```